NAME: M.Madhulika

REG NO: 192472188

Course: c++ programming DSA0108

1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a **Consultation Charge System** for hospital management.

Task:

a. Use **function overloading** to calculate charges for **general consultation** and **specialist consultation**.

b. Use **call by reference** to modify the final bill amount.

c. Use **default arguments** to add a standard service charge if needed.

d. Use **function prototyping** for all calculation functions.

e. Display the **patient type** and **total consultation charge**. **(BL3)**

```
#include<iostream>
using namespace std;
float charge(float g,float s=100);
float charge(float g,int sp,float s=100);
void bill(float &b);
float charge(float g,float s){return g+s;}
float charge(float g,int sp,float s){return g+sp+s;}
void bill(float &b){b*=1.05;}
int main(){
int ch,sp;
float fee,b;
  cin>>ch>>fee;
  if(ch==1){
    b=charge(fee);
    bill(b);
    cout<<"specialist\n:"<<b;}
  else{
    cin>>sp;
    b=charge(fee,sp);
```

1
500

```
  if(ch==1){
    b=charge(fee);
    bill(b);
    cout<<"specialist\n:"<<b;}
  else{
    cin>>sp;
    b=charge(fee,sp);
    bill(b);
    cout<<"specialist\n:"<<b;}}
```

OUTPUT

```
specialist
:630
```

2. Design a **C++ program for a Fitness Centre Membership System**. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Task:

a. Create a class **Member** with the attributes: **Member ID, Name, Plan Type, and Fee**.

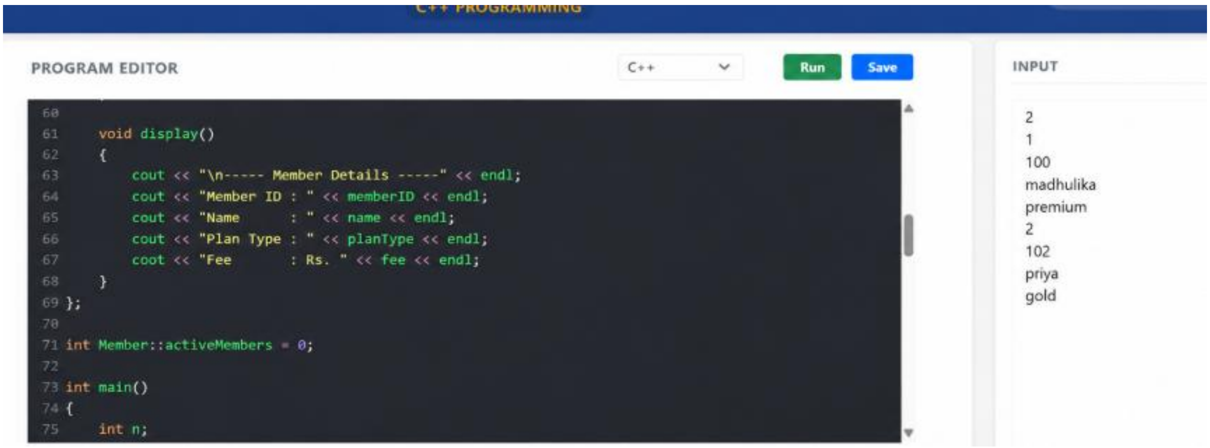
b. Implement member functions to **assign plans, calculate fees, and display membership details**.

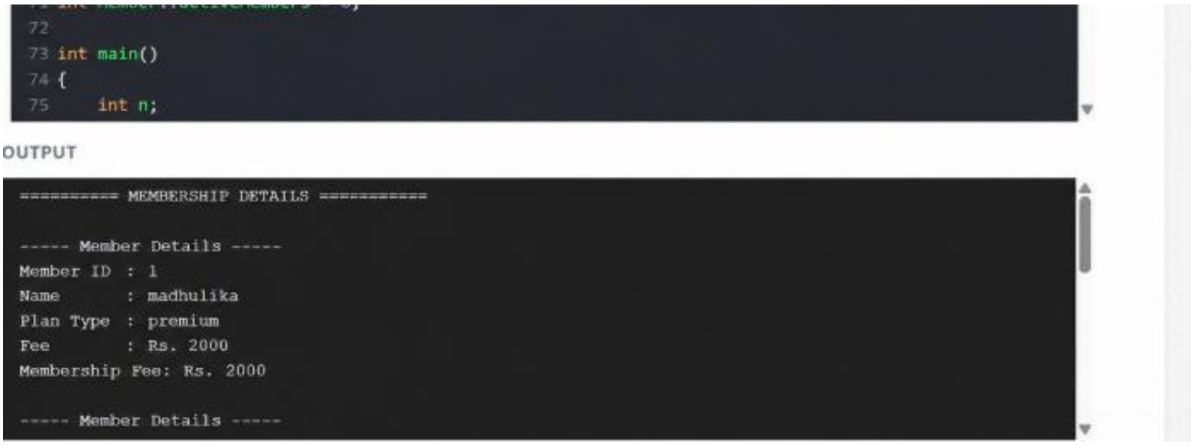
c. Use an **array of objects** to manage multiple members.

d. Add **validation** for valid membership plans.

e. Include a **static member** to track the total number of active members.

f. Evaluate how your design supports **easy modification of pricing plans**.





3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

Task:

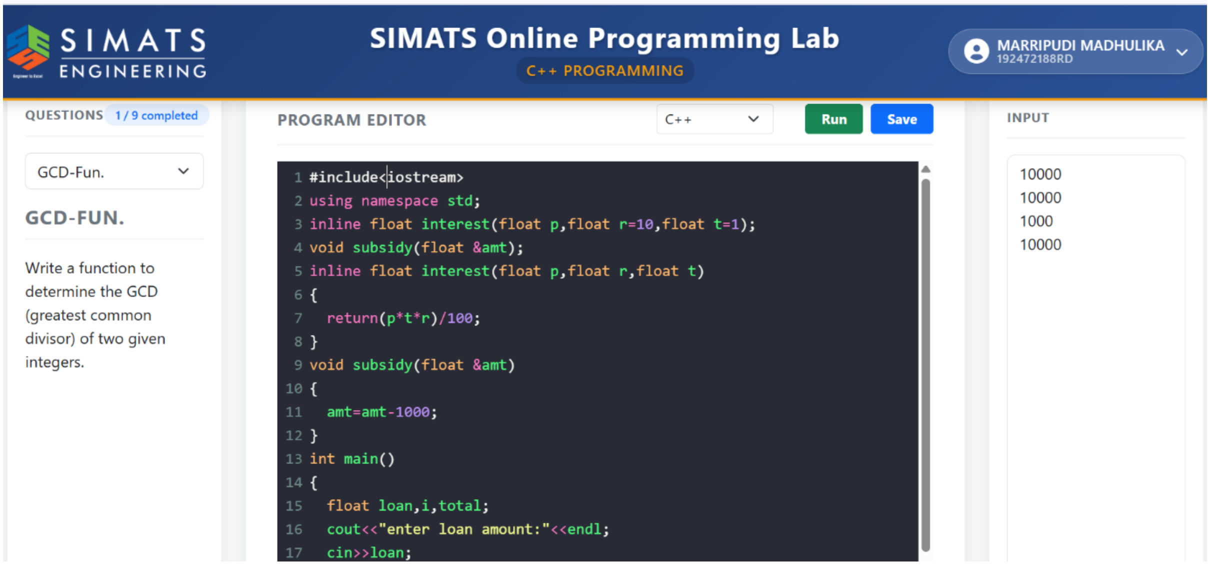
a. Use **function prototyping** to declare EMI-related functions.

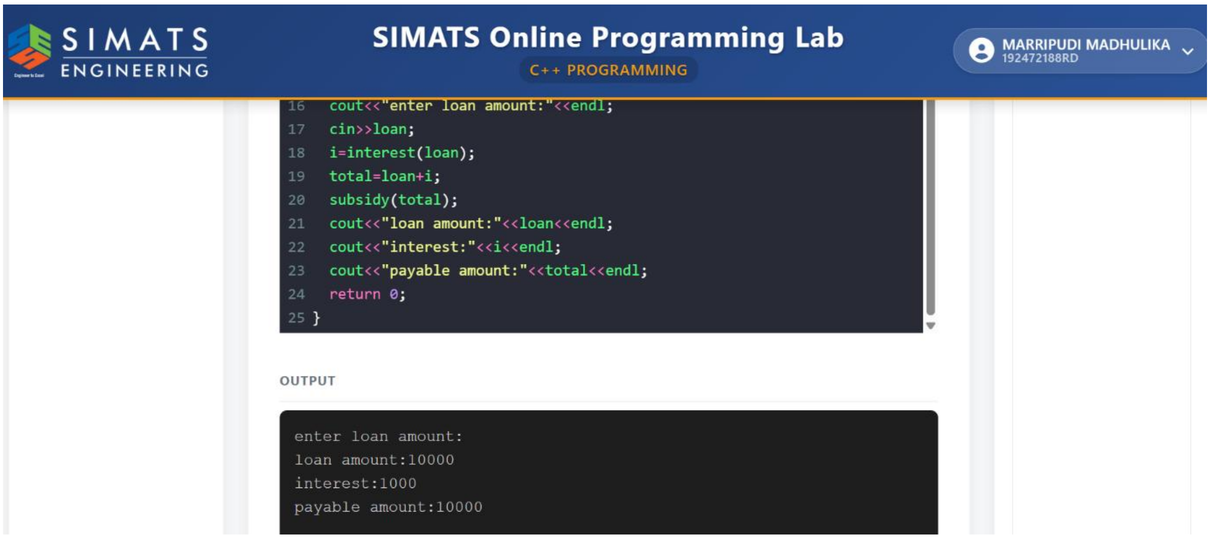
b. Use **default arguments** for the default interest rate where applicable.

c. Apply **call by reference** to update the payable amount after subsidy.

d. Use **inline functions** for simple interest calculations.

e. Display the **loan amount, interest, EMI, subsidy (if any), payable amount, and repayment details** clearly.





4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System.

a. Create a class Parcel with appropriate private data members such as parcel type, weight, distance, and packaging charge.

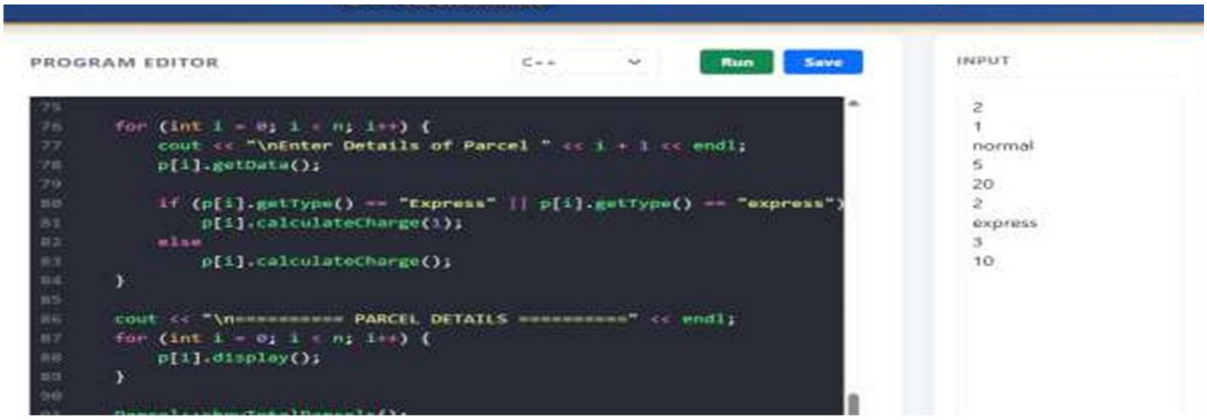
b. Use public member functions to read parcel details, calculate the delivery charge, and display the parcel information.

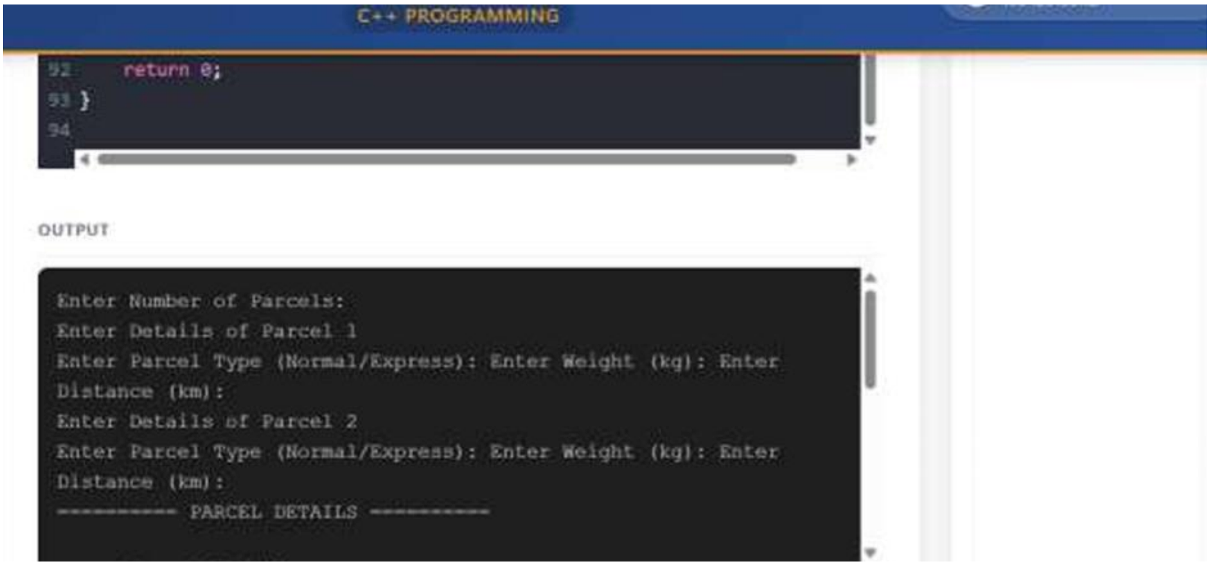
c. Use function overloading to calculate charges for Normal Delivery and Express Delivery.

d. Use default arguments to assign a default packaging charge when it is not specified by the user.

e. Use an inline function to calculate the base charge based on parcel weight and delivery distance.

f. Maintain the total number of parcels processed using a static data member and display it through a static member function.





5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions and object-oriented programming concepts to develop a **Water Billing System**.

a. Create a class **Consumer** with appropriate private data members such as **consumer ID, consumer type, water consumption (in liters), and bill amount**.

b. Use public member functions to **input consumer details, calculate the water bill, and display the billing information**.

c. Implement **function overloading** to calculate water bills for **Domestic** and **Commercial** consumers using different tariff rates.

d. Use **call by reference** to update the bill amount after applying a government subsidy.

e. Use **default arguments** to include a service tax, where a default tax rate is applied unless specified.

f. Declare all functions using **function prototyping** before their definitions.

g. Maintain the total number of consumers processed using a **static data member** and display it through a **static member function**.

h. Create an **array of objects** to store and process the details of multiple consumers, and display the **consumer type, water consumption, and subsidy-adjusted bill amount**.